

Land, Credit, Overvaluation, and Debt Rollover: A Systematic Review of the Stiglitz–Hirano–Toda Literature on Land Rent and Financialization*

WORK IN PROGRESS / PLEASE DO NOT CIRCULATE

Fondecyt N° 1262439

June 23, 2026

Abstract

This review presents an overview of the Stiglitz–Hirano–Toda (Stiglitz–Hirano–Toda) literature on the role of land in economic development. The authors’ central claim is that land is not a neutral asset. Rather, because it is non-reproducible, durable, scarce, and capable of performing multiple economic functions—as a factor of production, a store of value, and a collateral vehicle for credit expansion—land can reorganize macro-financial relations. Land-price inflation can raise measured wealth without expanding physical capital accumulation; sectoral credit allocation, when routed toward land and the real estate sector, can enlarge balance sheets while reducing long-run growth; unbalanced growth, understood as differential productivity growth between land and non-land factors, can decouple land prices from rent fundamentals; and land can modify the relationship between returns, growth, and debt rollover. The review identifies four mechanisms—capital-formation crowding-out, sectoral credit and collateral allocation, unbalanced-growth land overvaluation, and the land-mediated debt-rollover reasoning—and evaluates the theoretical underpinnings, strengths, and weaknesses of the Stiglitz–Hirano–Toda literature. Its concluding discussion translates these formal mechanisms into a geographical political economy framework, examining how land functions as High-Quality Collateral in endogenous cycles of financial fragility, distributive conflict, and structural asset inequality.

Keywords: Land Rent; Land Speculation; Financialization; Credit Allocation; Land Overvaluation; Debt Rollover; Unbalanced Growth; Physical Capital Formation.

JEL Codes: B51; E22; E44; G12; O11; P16; R30.

*This document is a working synthesis prepared for the Fondecyt project on land-rent financialization in Chile.

Review question

How does the Stiglitz–Hirano–Toda literature explain the relationship between land-price inflation, credit expansion, unbalanced growth, productive capital formation, land overvaluation, and debt rollover, and what can be carried into a Fondecyt project on Latin American land-rent financialization?

Executive Synthesis

The Stiglitz–Hirano–Toda literature approaches land not as a neutral background asset, but as a non-produced, durable, and inelastic real asset that functions as a production factor, a macro store of value, and, crucially, as High-Quality Collateral (HQC). Because land does not undergo economic depreciation, it underpins the endogenous creation of long-term financial liabilities—driving the “Great Mortgaging”—rather than functioning as a financial asset itself. This unique institutional role engenders macro-financial imbalances, structural stagnation, and escalating wealth inequality. When land functions primarily as a collateralized store of value, land-price inflation inflates measured balance-sheet wealth without expanding physical capital accumulation. Consequently, credit directed toward real estate expands institutional balance sheets while systematically retarding long-run growth through the defunding of productive investment.

The core insight of the Stiglitz–Hirano–Toda framework is that land valuation becomes macroeconomically dominant when credit systems, collateral metrics, and aggregate growth pathways organize around a non-reproducible asset. This systematic review synthesizes four core mechanisms within this literature: (i) the crowding-out of physical capital accumulation driven by speculative asset bidding; (ii) the sectoral misallocation of credit and asymmetric collateral rules that starve the productive sector; (iii) the structural decoupling of land prices from rent fundamentals under conditions of unbalanced growth ($\sigma > 1$); and (iv) the tendency of land-mediated asset absorption to alter the relationship between returns and growth ($R < G$), reopening conditions for infinite sovereign debt rollover.

While these mechanisms exhibit high formal validity, they remain institutionally thin due to their reliance on neoclassical overlapping-generations (OLG) models. To translate these mechanisms into the political economy of Latin American land-rent financialization, this review moves beyond the OLG fiction of “young agents buying from old agents.” Instead, it reinterprets the Stiglitz–Hirano–Toda architecture through a heterodox and Post-Keynesian lens: land financialization is analyzed as the balance-sheet management of financialized owners, developers, and states facing continuous rollover needs. Within this framework, endogenous Minskyan fragility and distributive cycles generate a persistent structural pressure toward asset inequality, even when aggregate macroeconomic dynamics appear bounded. Ultimately, the review demonstrates how the formal boundaries of the Stiglitz–Hirano–Toda corpus can be grounded in the concrete institutional realities of collateralized debt creation, spatial segregation, and the structural absence of collateral for the non-land-owning majority in the periphery.

1 Introduction

P01. Land-price inflation distorts macroeconomic dynamics precisely because it expands balance-sheet wealth without driving physical capital accumulation (Stiglitz, 2015a, pp. 1–2, 4, 11); (Stiglitz, 2015b, pp. 10–11, 28). Unlike machinery, industrial plant, or structures, land represents an unproduced, structurally inelastic locational claim. Consequently, land-price appreciation registers as private wealth expansion, amplified collateral values, or localized capital gains while leaving the real, produced capital stock unaltered. Stiglitz’s wealth-measurement framework establishes the accounting foundation for this decoupling: aggregate measured wealth can escalate through the capitalization of land and monopoly rents even when physical capital accumulation and national saving stagnate or contract in proportion. The core issue extends beyond statistical mismeasurement; it exposes a structural contradiction wherein an economy records intensive balance-sheet appreciation while its actual stock of produced means of production grows slowly or contracts relative to total assets.

P01b. To map the precise mechanics of this decoupling, Stiglitz’s canonical accounting equations serve as a foundational baseline for tracking the wealth-capital divergence (Stiglitz, 2015a,b,c). In standard neoclassical paradigms, aggregate wealth and productive capital are treated as effectively synonymous variables. Alternatively, Stiglitz breaks this conceptual identity by decomposing total balance-sheet measured wealth (W) into three structurally distinct components:

$$W = K + P_T T + V_B \quad (1)$$

P01c. In this identity, K represents the stock of reproducible, depreciating physical capital accumulation, $P_T T$ constitutes the capitalized value of inelastic locational claims (where P_T is the market price of land and T is the fixed physical supply), and V_B captures the pure speculative bubble or capitalized residual structural rents. This formalizes how financialization triggers an artificial expansion of W without inducing a corresponding expansion of K . When central bank monetary easing intersects with permissive commercial bank lending, credit expands non-neutrally, using land as a preferred collateral base (Stiglitz, 2015c). Because land is an unproduced and inelastic asset, this liquidity influx does not finance physical investment; instead, it is absorbed directly into the asset price ($P_T \uparrow$) and the speculative residual ($V_B \uparrow$). Consequently, the measured wealth-to-income ratio (W/Y) escalates alongside top-tail inequality, while the actual path of physical capital accumulation per worker remains flat, capping real wages and permanently stunting long-run economic growth.

P02. The organizing architecture of this systematic review is built around relational mechanisms rather than a serial paper-by-paper summary. The tension between land as a macro store-of-value vehicle and physical capital accumulation defines the primary set of inquiry. The interplay between credit expansion and asset pricing connects financial channels, while the variance between unbalanced growth dynamics and rent fundamentals isolates the formal parameter boundaries of land overvaluation. Overall, the interactions among asset returns (R), macroeconomic growth (G), and structural debt rollover allow the authors in this body of literature to establish a channel for debt-sustainability. Within this framework, wobbly dynamics remains strictly supplementary: it theorizes endogenous instability and phase transitions under volatile expectations, but it does not displace or override the four primary structural mechanisms (Hirano and Stiglitz, 2022a, pp. 7, 24,

46–47); (Hirano and Stiglitz, 2022b, pp. 1–2, 10).

P03. This systematic review maintains a strict analytical boundary between the internal mathematical modeling environments of the Stiglitz–Hirano–Toda literature and the extracted macro-financial mechanisms under examination. The modeling machinery of this framework—based on two-period overlapping-generations (OLG) micro-foundations, rational expectations trajectories, no-arbitrage asset-pricing profiles, and formal credit-friction parameters (θ, θ^x) —is treated as the baseline architecture of the corpus (Hirano and Stiglitz, 2024; Hirano and Toda, 2025b). This review deliberately limits its scope to the resulting causal relations of interest for political economy. By isolating these structural dynamics, the text brings to the fore the core distributive and allocation frictions—specifically how an inelastic locational claim operating as a macro store-of-value vehicle alters the long-run trajectory of physical capital accumulation—without absorbing the highly restrictive general equilibrium assumptions required to close the formal neoclassical macro-finance models.

2 Methodological Framework and Corpus Design

P04. This systematic review comprises a targeted corpus of fourteen core macro-financial working papers and peer-reviewed articles developed within the Stiglitz–Hirano–Toda (Stiglitz–Hirano–Toda) research program between 2015 and 2026. The corpus was selected based on its explicit formalization of land as a non-produced asset interacting with credit constraints, capital accumulation, and aggregate growth. To enforce analytical accountability and prevent generalized synthesis, data extraction was managed through a structured coding matrix. This matrix maps specific theoretical claims, parametric configurations, and policy variances directly to primary source citations, ensuring that every mechanism synthesized in this review is strictly grounded in the formal architecture of the original models rather than informal interpretation.

P05. The extraction protocol partitions the literature across five strict analytical coordinates to trace how a non-produced asset reorganizes macroeconomic balance sheets:

- **Definitions:** Isolating the structural boundaries between total balance-sheet wealth (W), reproducible physical capital (K), pure land price (P_t), and ground rent flows (D_t).
- **Mechanisms:** Tracking portfolio margins, specifically the general equilibrium crowding-out of physical capital accumulation driven by asset bidding.
- **Causality:** Outlining the direction of transmission channels, moving from structural leverage loosening ($\lambda > \bar{\lambda}$) or monetary expansion to land-price inflation.
- **Parameters:** Mapping the explicit locus of optimization, notably the elasticity of substitution ($\sigma > 1$) and asymmetric collateral capacities (θ, θ^x) .
- **Epistemological Status:** Explicitly separating abstract deductive proofs from their motivating advanced-economy empirical facts and transportability limits.

P06. The textual meta-synthesis is distributed across six operational clusters that structure the review’s progression. Cluster C1 (Foundations) anchors the wealth-capital residual decoupling; C2 maps expectation-driven global instabilities and wobbly dynamic paths; C3 isolates low safe

interest rates and credit-led siphoning; C4 deconstructs leverage thresholds, phase transitions, and asset bubbles; C5 re-evaluates the transversality conditions (R versus G) governing sovereign deficit rollover; and C6 establishes the transportability boundaries of the framework for peripheral political economy. By organizing the text along this matrix, the review surfaces the formal economic relations of interest while maintaining strict fidelity to the underlying mathematical architectures.

3 Conceptual and Analytical Foundations of Land Macro-Finance

P07. The macro-financial research program developed by [Hirano and Stiglitz \(2022a,b\)](#) is built upon a non-linear baseline that endogenously generates structural volatility, hysteresis, and state-space bifurcations under rational expectations. Utilizing a two-period overlapping-generations (OLG) structure, this architecture demonstrates that decentralized market clearings generate a pervasive multiplicity of momentary equilibria where self-fulfilling expectational corridors determine long-run paths. Within this wobbly container, tradeable land operates not as a neutral background factor of production, but as a structurally inelastic locational claim and a primary macro store-of-value vehicle that directly competes with physical capital accumulation for finite lifecycle savings. At the micro-intensive level, this asset-menu portfolio allocation is formalized by the clearing condition:

$$k_{t+1} + P_t = s_t(w_t + e) \quad (2)$$

where k_{t+1} represents next-period physical capital accumulation per capita, P_t is the current spot price of the land asset, s_t is the endogenous lifecycle saving rate, w_t is the real wage flow, and e represents the non-wage young endowment. Bullish expectations regarding future land prices—i.e., a self-fulfilling, rational belief profile where agents expect future asset appreciation ($P_{t+1} \uparrow$), lowering the future safe interest rate ($r_{t+1} \downarrow$) and maximizing the leveraged return on land speculation relative to manufacturing investments—drive an immediate asset boom ($P_t \uparrow$) that starves the real sector of physical equipment, contracting the residual available for physical capital accumulation ($k_{t+1} \downarrow$).

P08. These micro-intensive portfolio dynamics scale directly into aggregate macro balance sheets, establishing the core accounting decoupling between total measured wealth (W_t) and the accumulation of physical capital (K_t) ([Stiglitz, 2015b,a](#)). While physical capital accumulation (K_t) consists strictly of reproducible, depreciating tangible assets used in production ($Y_t = F(K_t, L_t)$), total measured wealth is a broader balance-sheet asset valuation defined by the identity:

$$W_t = K_t + P_{Tt}T + V_{Bt} \quad (3)$$

where $P_{Tt}T$ represents the capitalized value of unproduced locational claims (P_{Tt} being the macro price of land and T the fixed physical supply), and V_{Bt} constitutes the pure speculative bubble or residual structural rent component. Stiglitz formalizes this discrepancy through the concept of the *wealth residual*—the widening empirical divergence between observed increases in aggregate net worth and the smaller increments explained by physical capital accumulation and national saving. This residual surfaces a major macro-financial distortion: an economy can register substantial balance-sheet expansion ($W_t \uparrow$) driven entirely by the monetary capitalization of fixed asset positions and monopolistic rent claims ($P_{Tt}T + V_{Bt} \uparrow$), even as its physical capital stock stagnates or retrogresses.

P09. Reconciling this asset appreciation with localized rent fundamentals requires parsing the Land Overvaluation Theorem under conditions of unbalanced growth (Hirano and Toda, 2025b). Land rent (D_t) is a periodic dividend flow generated by a site’s spatial use, whereas the land price (P_t) represents the capitalized asset price of that location, calculated via a state price deflator (π_s) determined by the return margins of constrained investors:

$$V_t = \frac{1}{\pi_t} \mathbb{E}_t \sum_{s=t+1}^{\infty} \pi_s D_s \quad (4)$$

True overvaluation emerges as a necessary, unique long-run equilibrium outcome ($P_t > V_t$) when the elasticity of substitution between land and non-land factors exceeds one ($\sigma > 1$) and technological progress is faster in non-land sectors ($G > 1$). Under these parameters, the modern economy smoothly substitutes capital and labor for land (e.g., urban densification), causing the real share of output captured by productive ground rents to decay asymptotically toward zero ($\lim_{t \rightarrow \infty} \frac{r_t X}{Y_t} = 0$). Paradoxically, because land remains a macro store-of-value savings vehicle pulled upward by aggregate wage growth, its asset price separates permanently from its decaying rent fundamentals, bounded only by the down-payment constraints of constrained investors.

P10. This overvaluation cannot be detected through aggregate real estate indexes because housing and urban properties are composite commodities that join land to produced, depreciating structures. A misleading inference common to standard macroeconomics follows from this insight: a stationary aggregate housing rent-to-price ratio (D_t/P_t) does not preclude the existence of an under-the-structure land bubble (Toda, 2025). Because structural depreciation on building components continuously offsets asset inflation, housing data can appear empirically anchored while the pure, non-produced land component is undergoing explosive overvaluation. When the commercial banking system expands credit using real estate as its primary collateral base, this index opacity fuels a closed feedback loop: bank credit inflates the value of the non-produced site, higher valuations expand collateral buffers for credit-borrowing, unlocking additional land-directed credit volume, structurally diverting savings from physical capital accumulation towards non-productive assets.

4 Mechanism I: Land-Price Inflation and Capital-Formation Crowding-Out

P11. The primary transmission channel through which land speculation reorganizes macroeconomic paths operates via the direct general equilibrium de-funding of physical capital accumulation (Hirano and Stiglitz, 2022a, 2025b). Within the lifecycle saving space, an increase in the current spot price of land (P_t) represents an expanding financial claim on a finite national savings pool, mechanically contracting the remaining pool available for productive capital equipment (k_{t+1}) (Hirano and Stiglitz, 2022a). Because land functions as a non-produced, inelastic locational claim and a macro store-of-value vehicle rather than a reproducible asset, this price appreciation generates balance-sheet paper wealth gains without enhancing capital deepening (Stiglitz, 2015b; Hirano and Stiglitz, 2024). As land speculation intensifies, the general equilibrium crowding-out effect dominates the short-run partial equilibrium liquidity effects, resulting in a permanent structural stagnation of long-run growth:

$$\frac{d(1+g^*)}{d\theta^x} < 0 \quad (5)$$

where g^* represents the long-run economic growth rate and θ^x isolates the institutional leverage capacity attached to land collateral financing (Hirano and Stiglitz, 2024). By starving the productive sector, this speculative siphon stalls the learning-by-doing (LBD) externality loops that generate endogenous scale economies, trapping the macroeconomy in a persistent low-growth while generating top-end wealth concentration (Hirano and Stiglitz, 2024, 2025b).

P12. The dynamic transmission of crowding-out capital accumulation extends beyond balance-sheet accounting, instead, the literature portrays and models it as a self-reinforcing expectational feedback loop. A localized demand boom forces an immediate inflation in the spot price of the inelastic locational claim (P_t), driven by forward-looking agents who utilize the asset as a macro store-of-value vehicle whose demand is sustained by rational resale expectations ($P_{t+1} \uparrow$) (Hirano and Stiglitz, 2022a, p. 7). Because the pool of savings are finite, higher land prices automatically reduce current physical investment, contracting dynamically — i.e. to future temporal horizons — physical capital accumulation ($k_{t+1} \downarrow$) (Hirano and Stiglitz, 2022a, p. 16). This asset-driven siphon permanently depresses the trajectory of future real wages (w_{t+1}) and aggregate output (Y_{t+1}), yet in the context of a closed economy and a neo-classical production function with diminishing marginal returns, the physical capital scarcity rises its marginal product, pushing the real interest rate upward to maintain competitive conditions without arbitrage (Hirano and Stiglitz, 2022a, p. 24). To satisfy this moving margin, the unleveraged rate of return on land, $R_t^x = (D_{t+1} + P_{t+1})/P_t$, must appreciate at an accelerating velocity to remain competitive relative to the real interest rate (Hirano and Stiglitz, 2022a, p. 10). The speculative boom therefore carries an intrinsic, structural instability: it requires explosive, continuous price inflation to sustain its own attraction, even as the underlying asset absorption undercuts the physical capital base that generates the economy’s long-run lifecycle income (Hirano and Stiglitz, 2022a, p. 46). Once this divergent path breaches its calculation bounds, a phase transition occurs; expectations endogenously reverse from bullish to bearish ¹, and the speculative bubble collapses into an endogenous macroeconomic crash (Hirano and Stiglitz, 2022a, p. 47).

P13. The policy-theoretic counterfactual of this siphoning mechanism is formalized through a modern general equilibrium reconstruction of land rent taxation (Hirano and Stiglitz, 2025b, pp. 327, 329). Within a lifecycle framework where a non-produced asset in fixed supply competes directly with reproducible capital for a share of finite savings, a targeted tax on land rents or capital gains acts as an asset-allocation stabilizer (Hirano and Stiglitz, 2025b, p. 335). Rather than operating as a neutral revenue-generating instrument, land taxation structurally alters portfolio incentives by lowering the after-tax leveraged return of land speculation relative to industrial investment (Hirano and Stiglitz, 2025b, p. 345). By compressing the speculative premium of inelastic locational claims, this fiscal intervention effectively redirects the marginal savings envelope back toward the physical capital accumulation, thereby unlocking the learning-by-doing scale externalities that drive long-run

¹i.e., a self-fulfilling, rational reversal of the expectational profile where agents, recognizing that the explosive trajectory of the inelastic locational claim has breached its macroeconomic calculation bounds, endogenously shift from bullish to bearish sentiments; this immediate contraction in expected future land prices ($P_{t+1} \downarrow$) drives a collapse in current asset valuations ($P_t \downarrow$) and triggers a temporary overinvestment in physical capital accumulation as the remaining lifecycle savings pool is rapidly reallocated away from non-productive assets.

productivity growth enhancing aggregate consumption throughout the lifecycle of agents within a OLG neo-classical framework (Hirano and Stiglitz, 2025b, pp. 329, 349).

P14a. The long-run growth penalties established by the Stiglitz–Hirano–Todal literature ($\frac{d(1+g^*)}{d\theta^x} < 0$) offer a clear compositional policy lesson: macroeconomic stability depends less on aggregate credit volume than on its sectoral allocation (Hirano and Stiglitz, 2024). To prevent asset inflation from driving the systemic de-funding of the real economy, financial regulation must structurally restrict the banking system’s capacity to leverage non-produced assets (Hirano and Stiglitz, 2024). Specifically, central authorities must maintain regulatory collateral thresholds for real estate (θ^x) that are strictly tighter than the borrowing limits allocated to industrial investment (θ), thereby forcing credit lines into productive capital accumulation (Hirano and Stiglitz, 2025a). While the Stiglitz–Hirano–Todal literature notes that secondary fiscal tools—such as capital-gains taxes or land-rent taxation—can theoretically lower the attractiveness of speculative real estate portfolios, it cautions that under rational expectations, uncoordinated or uncredited tax interventions risk paradoxically increasing asset price volatility and accelerating capital flight (Hirano and Stiglitz, 2025b).

P14b. The set of regulations and fiscal policies toolkit suggested by the authors exposes a critical friction when evaluated from a geographical political economy standpoint. The translation of abstract collateral constraints ($\theta^x > \theta$) into functional urban policy assumes the existence of a cohesive, transparent state apparatus capable of penalizing ground-rent extraction without disrupting macro-financial liquidity. In peripheral institutional contexts, this regulatory capacity is systematically undermined by obsolete or highly fragmented municipal property registries, widespread informal land tenure systems, and deep-seated political capture by banking-developer syndicates. Consequently, attempting to operationalize the literature’s policy solutions without historicizing the local state’s dependence on real estate development risks reproducing spatial exclusion: tighter credit caps frequently block lower-income housing production while corporate land-banking blocks smoothly bypass macroprudential boundaries through offshore financial routing.

5 Mechanism II: Sectoral Credit and Asymmetric Collateral Allocation

P15. The second mechanism shifts the analytical focus from the total volume of credit to its sectoral allocation. Hirano and Stiglitz (2024) establish that long-run growth outcomes depend not on aggregate credit expansion, but on the specific asset classes toward which commercial bank loans are routed. When credit markets operate under institutional frictions, entrepreneurs must allocate available leverage between two competing destinations: the accumulation of physical capital (K_t) inside a productive sector, or the acquisition of an inelastic locational claim and macro store-of-value vehicle inside an unproductive real estate sector (Hirano and Stiglitz, 2024, 2025a). If credit expansions are disproportionately channeled into property speculation, a larger credit stock directly correlates with lower long-run growth, as the flow of new credit finances speculative land down payments or the acquisition of assets using land as collateral, rather than physical capital accumulation (Hirano and Stiglitz, 2024).

P16. This asymmetric allocation is accelerated by expansionary monetary policy and financial deregulation. Lowering the risk-free safe interest rate ($r^* \downarrow$) or relaxing real estate leverage rules directly compresses the down payment required to buy land, inflating its current asset price (P_t) and artificially raising the leveraged return on land speculation relative to physical capital accumulation (Hirano and Stiglitz, 2025a). While standard financial theory suggests that higher collateral values relax borrowing constraints and stimulate investment (a “crowding-in” effect), the Stiglitz–Hirano–Toda framework demonstrates that the systemic, economy-wide outcome runs in reverse (Hirano and Stiglitz, 2024, 2025a). The portfolio shift toward the non-produced asset intensifies land speculation, systematically starving the productive sector of investment liquidity and trapping the macroeconomy in a low-growth corridor characterized by an expanded consumption-to-GDP ratio and a permanently depressed rate of physical capital accumulation:

$$\frac{d(1+g^*)}{d\theta^x} < 0, \quad \frac{d\phi^*}{d\theta^x} > 0, \quad \frac{d(1+r^*)}{d\theta^x} < 0 \quad (6)$$

where θ^x isolates the institutional borrowing capacity or leverage limits attached to real estate collateral, ϕ^* tracks the relative size of land speculation, and g^* defines long-run productivity growth (Hirano and Stiglitz, 2024).

P17. This mechanism is strictly governed by the binding collateral constraints faced by entrepreneurs, who must finance investments using a fraction of their internal net worth as a down payment. The regulatory parameter θ^x dictates the maximum leverage allowed against land. When this constraint is binding, any relaxation in θ^x disproportionately expands borrowing capacity for real estate, triggering the crowding-out effect. Conversely, the mechanism has strict institutional limits: if financial regulation strictly caps θ^x (macroprudential limits), or if entrepreneurs lack the internal net worth to meet the required down payments, the land-speculation channel is structurally choked off. Thus, the diversion of credit is not an automatic market outcome, but a direct function of how institutional leverage limits interact with the distribution of internal wealth.

P18. Stiglitz’s foundational analysis of land and credit provides the institutional bridge to these later macro-financial models (Stiglitz, 2015c). In this framework, credit availability and bank lending decisions drive land prices upward, raising measured balance-sheet wealth without increasing the equilibrium stock of physical capital. Because land serves as the primary collateral object, credit conditions directly inflate that collateral value, creating a circular feedback loop: credit supports land prices, higher land prices expand collateral buffers, and expanded collateral buffers unlock additional land-directed credit volume. Crucially, this feedback loop is inherently distributional. The expansion of collateral buffers accrues almost exclusively to existing landowners and real estate developers, whose balance sheets swell with unearned capital gains. This concentrated wealth is then recycled into further land speculation, while the crowding-out of physical capital suppresses the real wages of the non-land-owning majority, structurally widening the wealth gap.

P19. Consequently, this financial channel renders monetary policy inherently ambiguous. In standard macroeconomic theory, lower interest rates or higher liquidity are assumed to push investors from cash into productive capital. In the Stiglitz–Hirano–Toda framework, however, the available investment options include land, which can absorb the very liquidity intended for physical capital accumulation. Expansionary monetary policy can therefore either support or retard long-run growth,

depending entirely on whether it relaxes credit for productive investment or merely inflates land-collateral values. The macroeconomic outcome thus hinges on a compositional test: which sectors receive leverage, and which asset prices respond first to the injection of liquidity (Hirano and Stiglitz, 2025a, 2024).

P19b. Viewed through the lens of geographical political economy and heterodox macroeconomics, this abstract mechanism of asymmetric collateral allocation reveals a profound spatial and distributive reality. The Stiglitz–Hirano–Toda framework demonstrates formally that credit is not a neutral medium of exchange, but a structurally biased force that actively constructs a rentier economy. When the financial system disproportionately leverages land over physical capital, it is not merely “misallocating” resources in a frictionless vacuum; it is actively financing the enclosure, financialization, and spatial segregation of urban territories. The collateral feedback loop translates directly into a geography of exclusion: existing asset holders in prime locations capture unearned windfalls, while the crowding-out of productive investment suppresses aggregate wage growth and starves peripheral zones of productive infrastructure. Ultimately, lesson is that land financialization is not just a macroeconomic distortion to be corrected by optimal collateral rules; it is a territorial regime of accumulation where endogenous credit creation becomes the primary engine of spatial inequality and rentier power.

6 Mechanism III: Unbalanced Growth and the Land Overvaluation Theorem

P20. In the Stiglitz–Hirano–Toda branch, “unbalanced growth” carries a strict, formal neoclassical meaning: it refers to differential technological productivity growth between land and non-land factors. Specifically, labor or human-capital productivity grows faster than land factor capability ($G > 1$). It is crucial to distinguish this formal technological condition from the broader political economy concept of “disproportional growth” or “overaccumulation” developed in the project’s framework, which refers to non-proportional movements among demand, credit, land valuation, and productive capacity. While Hirano and Stiglitz separately model *credit misallocation*—demonstrating how financial liberalization diverts credit from productive capital into unproductive land speculation—the Land Overvaluation Theorem developed by Hirano and Toda isolates a deeper, purely technological driver. In that specific framework, the authors deliberately abstract from credit constraints to prove that differential factor productivity alone structurally sets the stage for financial distortion. Under these conditions, land rents grow slowly relative to wages and aggregate income, while land remains a durable store of value purchased out of rising savings (Hirano and Toda, 2025b).

P21. The Land Overvaluation Theorem formalizes how this differential productivity leads to a structural decoupling of land prices from rent fundamentals. The central mathematical condition requires the elasticity of substitution between land and non-land factors to exceed one ($\sigma > 1$). The inquiry regarding how elasticity relates to factor productivity is resolved by the model’s structure: while σ measures the *technological flexibility* to substitute factors in response to changing relative prices or costs, it is the *differential productivity growth* that actually drives those relative prices apart over time. Because $\sigma > 1$, the exponent in the convergence condition $\left(\frac{A_H}{A_X}\right)^{\frac{1}{\sigma}-1}$ becomes strictly negative (since $\frac{1}{\sigma} < 1$). This negative exponent ensures that the real share of output captured by

productive ground rents collapses asymptotically toward zero ($\lim_{t \rightarrow \infty} \frac{r_t X}{Y_t} = 0$). This reconciles with classical Ricardian rent theory: at the marginal site level, land still exhibits diminishing marginal returns; however, at the aggregate macroeconomic level, the high elasticity of substitution allows the modern economy to structurally bypass land bottlenecks (e.g., through urban densification), causing the aggregate rent share to decay even as the asset price inflates (Hirano and Toda, 2025b).

P22. Land overvaluation in this framework is not simply irrational exuberance; rather, it emerges as a structural, asymptotic tendency on a fundamental growth path where the asset price permanently separates from the present value of land rents. Crucially, this long-run decoupling does not imply a single, stable global steady state. As the wobbly dynamics literature demonstrates, the introduction of self-fulfilling expectations and leverage thresholds generates a multiplicity of momentary equilibria. Consequently, the economy may never smoothly converge to this fundamental overvaluation path; instead, it can become trapped in chaotic, non-convergent corridors or undergo endogenous phase transitions into outright speculative bubbles. It also differs fundamentally from a generic housing-price boom. Because housing is a composite commodity joining non-produced land to depreciating structures, an observed stationary aggregate housing rent-to-price ratio does not rule out a pure land bubble. The analyst must separate the rent on structures from the pure land dividend; otherwise, evidence against a housing bubble can be misread as evidence against land overvaluation (Toda, 2025).

P23. The Leverage Accelerator mechanism, developed by Hirano et al. (2024), explicitly reintroduces the credit misallocation dynamics abstracted away in the Overvaluation Theorem, deriving its financial transmission directly from those structural boundaries. While the Overvaluation Theorem defines the *static* long-run condition under which land prices *can* decouple from rents ($\sigma > 1$, differential productivity), the Leverage Accelerator models the *dynamic financial realization* that actually crosses this boundary. The crucial object is a critical institutional leverage threshold ($\bar{\lambda}$). Below this threshold, land prices track fundamental rent values. Above it, financial loosening activates a multiplicative accelerator loop: rising land prices expand collateral values, which relaxes borrowing constraints, funneling more credit back into land speculation. This produces a phase transition from a fundamental growth path to a nonstationary, trend-bubbly unbalanced path where the ratio of fundamental value to price approaches zero ($V_t/P_t \rightarrow 0$) (Hirano et al., 2024).

P24. The synthesis of these mechanisms is heterogeneous, and the effect of land-price inflation depends entirely on the specific model architecture. One branch shows land speculation crowding out physical capital accumulation by absorbing finite savings. Another branch shows leverage and land-price increases reinforcing asset demand up to a phase transition. A serious application to land-rent financialization cannot cite “the literature” as if it made one uniform claim; rather, it must trace whether land is absorbing savings away from capital, whether collateral values are relaxing investment constraints, and whether the equilibrium is fundamentally anchored or bubbly.

P24b. Viewed through the lens of geographical political economy and heterodox macroeconomics, the Stiglitz–Hirano–Toda mechanism of unbalanced growth and land overvaluation reveals a profound socio-spatial transformation. When differential productivity and high elasticity of substitution between standard factors of production (capital and labor) and non-reproducible land ($\sigma > 1$), drive the aggregate rent share asymptotically toward zero, land is effectively stripped of its classical role as

a productive input and is entirely subsumed as a pure financial asset. This structural “overvaluation” is not merely a pricing anomaly; it represents the financialization of space itself with land fulfilling a function as store of value. Urban land is no longer priced by its marginal productive contribution, but by its capacity to act as a spatial fix—a durable, inelastic sink for absorbing surplus liquidity and hoarding wealth. Consequently, the decoupling of land prices from rent fundamentals translates directly into a rentier geography, where the value of spatial assets might become detached from the real productive economy, structurally reinforcing spatial segregation, wealth concentration, and the power of asset holders over the built environment.

7 Mechanism IV: R versus G , Land, and Infinite Debt Rollover

P25. The R -versus- G framework addresses the fundamental macroeconomic question of debt sustainability, where R represents the safe real interest rate (or the return on assets) and G represents the aggregate economic growth rate. In conventional overlapping-generations (OLG) models without land, a condition where $R < G$ implies dynamic inefficiency due to oversaving, which theoretically allows a government to roll over its primary deficits infinitely (a Ponzi scheme). However, the standard neoclassical consensus dictates that introducing a productive, non-reproducible, and non-depreciating asset like land forces $R > G$ along a balanced growth path. Because land pays a positive productive rent, its total return (rent plus price appreciation) exceeds the growth rate, thereby pushing the safe interest rate above growth and ruling out infinite debt rollover (Hirano and Toda, 2025a).

P26. The Stiglitz–Hirano–Toda branch fundamentally disrupts this consensus by introducing the condition of unbalanced growth. Hirano and Toda (2025a) demonstrate that the conventional $R > G$ result relies on knife-edge balanced-growth restrictions. Once the model allows for unbalanced growth—where labor productivity grows faster than land productivity and the elasticity of substitution exceeds one ($\sigma > 1$)—the productive contribution of land becomes asymptotically negligible. Land rents decay toward zero relative to aggregate output, even as land remains a critical store of value for savings. Under these structural conditions, the dynamic relationship flips asymptotically back to $R < G$. This reopening of the $R < G$ condition within a Pareto-efficient land economy proves that infinite debt rollover remains possible, fundamentally altering the boundaries of fiscal sustainability.

P27. A subsequent iteration of this research explicitly cautions against treating $R < G$ as a naive policy shortcut (Hirano and Toda, 2026). The authors warn that market frictions generating $R < G$ do not automatically settle the debt-rollover question in a vacuum. The inequality is not self-interpreting; it depends entirely on which specific asset return is being compared to which growth rate, whether land is priced by the safe rate or by a constrained asset return, and whether the equilibrium is fundamental or bubbly. This caution is vital for understanding liquidity traps in financialized economies: when massive monetary liquidity (driving R down) is channeled disproportionately into land speculation, it creates a stagnation environment. The economy experiences high asset yields and inflated collateral values, but the real growth rate (G) collapses because productive capital is crowded out. Thus, high liquidity coexists with low growth, trapping the macroeconomy in a financialized stagnation corridor (Hirano and Toda, 2026; Hirano and Stiglitz, 2024).

P27b. Crucially, the reopening of the $R < G$ condition under unbalanced growth creates a unique fiscal space for what the literature terms “public Ponzi deficits.” Because the safe interest rate is structurally suppressed below the growth rate, the government can issue public debt and roll over its primary deficits indefinitely without the debt-to-GDP ratio exploding. In the Stiglitz–Hirano–Toda framework, this public debt acts as a competitive macroeconomic asset to the land bubble. By issuing public bonds, the government injects an alternative, safe store of value that directly competes with inflated real estate for finite lifecycle savings. This “bubble competition” serves as a powerful macro-stabilizer: it absorbs excess liquidity that would otherwise purely fuel speculative land overvaluation, thereby pushing back extreme wealth concentration in land assets. Consequently, continuous government deficit spending—far from being a mere fiscal failure—can structurally steer asset allocation, suppress the most destructive forms of land financialization, and enhance long-run productive growth and debt sustainability (Hirano and Toda, 2025a, 2026).

P28. This mechanism is central to the Fondecyt project’s core inquiry. The formal result concerning infinite debt rollover in OLG economies provides the exact macroeconomic transmission channel for understanding financial fragility in peripheral capitalist spaces. In Latin America, debt rollover—whether sovereign, corporate, or household—is the primary vulnerability of the economic structure. By demonstrating how land absorption of savings and the structural decoupling of land prices from rent fundamentals can alter the R versus G relationship, the Stiglitz–Hirano–Toda literature provides a rigorous theoretical architecture to explain how land-rent financialization structurally compromises long-term debt sustainability and macroeconomic stability.

P29. To operationalize this, it is imperative to clarify what the Stiglitz–Hirano–Toda literature understands by “land productivity.” In this framework, land is only “productive” in the classical sense—generating a marginal ground rent—insofar as it functions as a physical input in the production process. However, the Land Overvaluation Theorem demonstrates that under conditions of unbalanced growth—where labor and technological productivity grow faster than land capability—combined with an elasticity of substitution strictly greater than one ($\sigma > 1$), land undergoes a functional mutation. The condition $\sigma > 1$ is mathematically key because it represents the technological flexibility to substitute capital and labor for fixed land. As the productive sectors expand, this high substitutability allows the economy to structurally bypass land bottlenecks, causing the aggregate share of output captured by physical ground rent to decay asymptotically toward zero. Consequently, land ceases to be a productive input, mutating entirely into a pure financial macro store-of-value and collateral vehicle. Therefore, when land becomes a store of value and financial asset, its price is no longer determined by its “productivity” (its marginal capacity to generate rent), but entirely by its financial attributes: its capacity to hoard wealth, serve as collateral, and absorb liquidity. Recognizing this mutation is essential to avoid conflating the physical productivity of land with its financialized asset price.

P30. Viewed through the lens of geographical political economy and heterodox macroeconomics, the abstract R -versus- G condition and the OLG modeling of debt rollover must be translated into the concrete institutional realities of financialized territories. The formal mechanism of “young agents selling to old agents” to sustain asset prices is, in the real world of Latin American urban political economy, the mechanism of *financialized owners, developers, and states facing rollover needs*. Debt sustainability is not merely a macroeconomic inequality; it is the daily balance-sheet management of a rentier class that relies on continuous land-price inflation to refinance liabilities, roll over debt,

and avoid liquidation. When the structural conditions of unbalanced growth and land overvaluation trap the economy in a low- G environment, the financialized owners' reliance on continuous collateral expansion becomes a source of endogenous systemic fragility. Thus, the Stiglitz–Hirano–Toda macroeconomic mechanism translates directly into a heterodox theory of financial instability, where land-rent financialization is the primary engine of debt-dependent accumulation and spatial crisis.

8 Policy implications

Asymmetric Collateral and Sectoral Credit Allocation. The primary policy implication of the Stiglitz–Hirano–Toda corpus is not a blanket restriction on aggregate credit, but a strict compositional regulation of its sectoral allocation (Hirano and Stiglitz, 2024). Because financial liberalization that relaxes land-collateral constraints (θ^x) triggers a multiplicative accelerator loop that crowds out physical capital accumulation, macroprudential policy must legally separate the borrowing limits tied to productive capital (θ) from those tied to land (θ^x). Authorities must enforce strictly tighter leverage limits and down-payment requirements on real estate, structurally choking off the credit-fed phase transitions that drive land overvaluation. Furthermore, in peripheral economies, this mechanism is compounded by the *absence of collateral* for the majority: while the elite leverage inflated land values to extract rents, non-land-owning populations are pushed into informal, high-cost credit markets, structurally widening the wealth gap and deepening spatial segregation.

Land Taxation and the Defunding of Speculation. To correct the portfolio distortion where a non-produced asset competes with reproducible capital for finite savings, fiscal policy must deploy targeted land-rent or speculative capital-gains taxation (Hirano and Stiglitz, 2025b). Rather than operating merely as a redistributive revenue tool, land taxation acts as an asset-allocation stabilizer. By compressing the after-tax leveraged return on land speculation, the state redirects the marginal savings envelope back toward physical capital accumulation. This directly addresses the core concern of the project: it counteracts the *desfinanciamiento* (defunding) of capital accumulation caused by land overvaluation, reactivating the learning-by-doing externalities that drive long-run aggregate productivity and wage growth.

Compositional Monetary Policy and Liquidity Traps. In an economy where land functions as a primary macro store-of-value, standard expansionary monetary policy is inherently ambiguous (Hirano and Stiglitz, 2025a). Injecting aggregate liquidity or lowering the risk-free rate does not automatically fund physical capital accumulation; instead, it is frequently absorbed by inelastic land assets, inflating collateral values while crowding out productive investment. Central banks must therefore abandon aggregate monetary neutrality and adopt compositional liquidity targeting. Policy must be evaluated by monitoring which asset prices respond first to monetary injections, ensuring that credit channels are structurally directed toward manufacturing and physical capital rather than land-collateral inflation, thereby avoiding a financialized stagnation trap characterized by high liquidity and low growth.

Sovereign Debt as a Macro-Stabilizer. Under conditions of unbalanced growth where the safe interest rate is structurally suppressed below the growth rate ($R < G$), the Stiglitz–Hirano–Toda literature reveals a unique fiscal space for what can be termed “public Ponzi deficits” (Hirano and Toda, 2025a, 2026). By issuing safe public bonds, the government injects an alternative, risk-free

store of value that directly competes with inflated real estate for finite lifecycle savings. This “bubble competition” acts as a powerful macro-stabilizer: it absorbs the excess liquidity that would otherwise purely fuel speculative land overvaluation, effectively starving the land bubble and suppressing extreme wealth concentration in spatial assets. Thus, continuous, domestically absorbed government deficit spending can structurally steer asset allocation away from unproductive rentier hoarding.

Disaggregated Statistical Infrastructure. Macroeconomic sustainability cannot be tracked using aggregate balance-sheet wealth (W), which conflates physical capital (K) with capitalized land rents and speculative bubbles (V_B) (Stiglitz, 2015a). Following the wealth residual framework and Toda’s measurement warning, national accounts must strictly decompose composite housing assets into depreciating structures and pure, non-produced land (Toda, 2025). Failing to separate the land dividend from structural rents allows hidden, explosive land bubbles to masquerade as stable housing markets, blinding regulators to impending endogenous phase transitions and masking the structural regression of the actual productive capital stock.

The Closed-Economy Bias and Limits in Small Open Economies. A critical limitation of these policy prescriptions is that the Stiglitz–Hirano–Toda framework is overwhelmingly calibrated for closed, mature capitalist economies. In small open, developing economies, domestic financial variables are deeply subordinated to the global financial cycle. Capital mobility, foreign currency denomination of liabilities, and external sovereign risk premia mean that domestic interest rates are not purely endogenous. Attempting to use domestic sovereign debt to “compete” with a land bubble can easily trigger currency depreciation or external balance-of-payments crises. Similarly, domestic collateral caps can be bypassed by foreign direct investment, offshore corporate borrowing, or dollarized real estate transactions. Therefore, the SHT policy prescriptions face strict limits in the periphery: macro-financial regulation cannot rely solely on domestic credit allocation, but must actively manage external vulnerabilities and the structural power of global liquidity.

Institutional Translation for Peripheral Capitalist Spaces. Finally, the Stiglitz–Hirano–Toda policy toolkit assumes a cohesive, transparent state apparatus capable of penalizing ground-rent extraction without disrupting macro-financial liquidity. In Latin America, operationalizing asymmetric collateral caps or land taxation requires confronting deep-seated geographical and institutional frictions: obsolete municipal cadastres, widespread informal land tenure, dollarized property markets, and political capture by banking-developer syndicates. Attempting to apply abstract macroprudential rules without historicizing the local state’s dependence on real estate development risks severe spatial backlash. Tighter credit caps frequently block lower-income housing production, while corporate land-banking smoothly bypasses regulatory boundaries through offshore routing. Consequently, macro-financial regulation in the developing world must be inextricably coupled with spatial, cadastral, and institutional reforms to be effective.

9 Discussion: Land, Collateral, and the Political Economy of Financialized Accumulation

The Institutional Role of Land: Collateral versus Financial Asset. While the Stiglitz–Hirano–Toda corpus effectively models land as a macroeconomic store of value that absorbs savings and anchors collateral, treating it analytically as a “financial asset” can obscure its specific institutional

and accounting role. In portfolio allocation frameworks, land naturally behaves like a financial asset—yielding a return, storing value, and competing with physical capital for investment. However, from an institutional accounting perspective, land ownership does not generate a corresponding financial liability for another economic agent; it remains a net real asset. Its distinct macroeconomic role derives not from being a financial claim, but from its physical non-depreciation and spatial fixity. These properties allow land to function as High-Quality Collateral (HQC), serving as the foundational real asset that underpins the expansion of long-term credit. Consequently, rather than conceptualizing financialization strictly as the transformation of land *into* a financial asset, it is analytically more precise to view land as the non-depreciable anchor that enables the endogenous creation of financial liabilities. This institutional refinement shifts the analytical focus from abstract portfolio allocation to the structural fragility of collateralized debt hierarchies and the broader dynamics of the “Great Mortgaging.”

Endogenous Instability: A Minskyan Translation of “Wobbly Dynamics”. The Stiglitz–Hirano–Toda literature introduces “wobbly dynamics” and endogenous phase transitions to explain why land-price paths do not converge smoothly to a steady state. While mathematically rigorous, this concept requires an institutional translation to capture the systemic nature of financial crises. Viewed through the lens of Minsky’s Financial Instability Hypothesis, the Stiglitz–Hirano–Toda leverage threshold ($\bar{\lambda}$) can be reinterpreted as the boundary between hedge, speculative, and Ponzi finance. Financial fragility is not an exogenous shock but an endogenous outcome of the accumulation process itself. As collateral values inflate and leverage expands, the resulting debt commitments alter the distributive conflict. Drawing on the tradition of distributive conflict models, high leverage introduces a debt-contingent wage discipline: when financial obligations cross critical thresholds, they suppress aggregate demand and reshape the Phillips curve. Thus, the “wobbly” phase transitions observed in formal models manifest in reality as Minskyan moments, where the internal dynamics of collateralized credit creation inevitably generate systemic fragility.

Distributive Cycles and the Structural Pressure Toward Asset Inequality. While the Stiglitz–Hirano–Toda mechanisms demonstrate how land financialization crowds out physical capital, they do not fully capture the cumulative, long-run distributional outcomes of this process. A Post-Keynesian extension reveals that asset inequality is not merely a byproduct of exogenous shocks, but an induced outcome of endogenous distributive and financial cycles. Even when aggregate macroeconomic dynamics remain bounded—exhibiting stable, cyclical fluctuations in employment and output—the financialization of land generates a persistent structural pressure toward wealth concentration. This occurs through the emergence of a “financial outside option.” As land values inflate, the opportunity cost of holding physical capital rises, diverting profit claims into financial and rentier income. This creates a financial pressure index that systematically outpaces macroeconomic growth. Consequently, the same local stability boundaries that govern macroeconomic cycles also partition regimes by the persistence of financial pressure, ensuring that asset-income claims accumulate disproportionately at the top, even in the absence of overt macroeconomic crises.

From OLG Fictions to Peripheral Political Economy. Ultimately, the translation of the Stiglitz–Hirano–Toda framework into the context of Latin American land-rent financialization requires discarding the OLG ontology of “young agents buying from old agents.” In the periphery, the circulation of land assets is driven by the balance-sheet management of financialized owners,

real estate developers, and states facing continuous rollover needs. The Stiglitz–Hirano–Toda corpus provides the formal boundaries of these mechanisms—demonstrating how unbalanced growth, sectoral credit misallocation, and collateral accelerators operate. However, the actual engine of financialization in developing economies must be analyzed through the institutional realities of High-Quality Collateral, endogenous Minskyan fragility, and the structural pressure of distributive cycles. By integrating the formal precision of the Stiglitz–Hirano–Toda mechanisms with a heterodox understanding of collateralized debt creation and asset inequality, this research agenda establishes a robust foundation for analyzing the spatial and macroeconomic crises of financialized land in the Global South.

References

- Hirano, T., Jinnai, R., and Toda, A. A. (2024). Leverage, endogenous unbalanced growth, and asset price bubbles. arXiv:2211.13100v7 [econ.TH], 13 Feb 2024; manuscript dated February 15, 2024.
- Hirano, T. and Stiglitz, J. E. (2022a). Land speculation and wobbly dynamics with endogenous phase transitions. NBER Working Paper 29745, National Bureau of Economic Research, Cambridge, MA.
- Hirano, T. and Stiglitz, J. E. (2022b). The wobbly economy: Global dynamics with phase and state transitions. NBER Working Paper 29806, National Bureau of Economic Research, Cambridge, MA.
- Hirano, T. and Stiglitz, J. E. (2024). Credit, land speculation, and long-run economic growth. NBER Working Paper 32479, National Bureau of Economic Research, Cambridge, MA.
- Hirano, T. and Stiglitz, J. E. (2025a). Credit, land speculation, and low-interest-rate policy. Local manuscript version dated August 21, 2025; arXiv first posted March 30, 2025.
- Hirano, T. and Stiglitz, J. E. (2025b). Henry george, land speculation, and economic growth and transformation. *Oxford Review of Economic Policy*, 41(2):326–357.
- Hirano, T. and Toda, A. A. (2025a). Land and infinite debt rollover. arXiv:2508.16002v2 [econ.TH], 16 Sep 2025; manuscript dated September 18, 2025.
- Hirano, T. and Toda, A. A. (2025b). Unbalanced growth and land overvaluation. CIGS Working Paper Series 25-011E, The Canon Institute for Global Studies. Forthcoming at Proceedings of the National Academy of Sciences.
- Hirano, T. and Toda, A. A. (2026). Land, G versus R, and infinite debt rollover. CIGS Working Paper Series 26-002E, The Canon Institute for Global Studies. January 15, 2026.
- Stiglitz, J. E. (2015a). The measurement of wealth: Recessions, sustainability and inequality. NBER Working Paper 21327, National Bureau of Economic Research, Cambridge, MA.
- Stiglitz, J. E. (2015b). New theoretical perspectives on the distribution of income and wealth among individuals: Part I. the wealth residual. NBER Working Paper 21189, National Bureau of Economic Research, Cambridge, MA.
- Stiglitz, J. E. (2015c). New theoretical perspectives on the distribution of income and wealth among individuals: Part IV. land and credit. NBER Working Paper 21192, National Bureau of Economic Research, Cambridge, MA.
- Toda, A. A. (2025). Land bubbles despite non-vanishing rents. *Economics Letters*, 257:112708.